

Credit Risk Decision Engine

End-to-End Machine Learning Project Report

Purpose: Build a credit-risk system that converts customer information into calibrated probability of default, operational lending decisions and portfolio-level expected-loss insights.

Frozen results: This report documents the completed Project 1 without changing the model, preprocessing, calibration, thresholds or final evaluation results.

Executive Summary

The project builds an end-to-end credit-risk decision engine using a LightGBM classifier followed by probability calibration. Calibrated probability of default (PD) is converted into APPROVE, REVIEW and REJECT actions and four risk bands. The system also connects model predictions to expected financial loss using $PD \times Exposure \times LGD$ and includes a reproducible deployment pipeline.

The dataset contains 172,205 training customers and 43,052 held-out test customers. A 106-feature raw/engineered input pipeline produces 226 model features. The final model records 0.7395 test ROC-AUC, 0.2264 test PR-AUC and 0.7340 ± 0.0018 OOF ROC-AUC. High and Very High Risk customers represent 28.29% of customers but 55.96% of expected loss.

1. Business Problem

A useful credit-risk system must do more than classify customers. It should estimate default probability, provide probabilities suitable for risk decisions, translate those probabilities into lending actions, segment customers by risk and quantify potential financial loss. This project implements that complete decision flow.

- Predict probability of default for each customer.
- Calibrate the predicted probabilities.
- Map calibrated PD to APPROVE, REVIEW or REJECT.
- Create four operational risk bands.
- Estimate portfolio expected loss and concentration.
- Validate the complete system for deployment and edge cases.

2. Dataset and Feature Pipeline

Item	Final value
Training customers	172,205
Held-out test customers	43,052
Raw/engineered inputs	106
Processed model features	226
Observed test default rate	8.11%

Feature engineering includes financial relationships such as credit-to-income, annuity-to-income, credit-to-goods, income per family member and credit per family member. Numeric and categorical inputs are transformed through the verified preprocessing pipeline. The reconstructed preprocessor matched the original processed matrices with zero numerical difference.

3. Model Development

The final predictive model is LightGBM. It generates raw default probabilities from 226 processed features. A separate calibration model then converts the raw probabilities into calibrated PD values for downstream risk decisions.

Metric	Result
Model	LightGBM + Calibration
Test ROC-AUC	0.7395
Test PR-AUC	0.2264
Calibrated Brier Score	0.0691
Train ROC-AUC	0.8571
Train-test gap	0.1176

4. Validation and Generalization

Evaluation uses both a held-out test set and 5-fold out-of-fold validation. OOF ROC-AUC is 0.7340 with standard deviation 0.0018, indicating stable fold-level discrimination. The 0.1176 train-test gap is retained as a limitation because it indicates remaining overfitting.

Validation measure	Result
Held-out ROC-AUC	0.7395
Held-out PR-AUC	0.2264
OOF ROC-AUC	0.7340
OOF standard deviation	0.0018
Train ROC-AUC	0.8571
Train-test gap	0.1176

5. Probability Calibration

Raw LightGBM probabilities are passed to a one-feature calibration model. The calibrated probabilities are used for business decisions rather than the raw model scores. The final calibrated Brier Score is 0.0691. Deployment validation explicitly checks that the calibrator receives one raw-probability feature.

6. Decision Engine

Decision	PD rule	Customer share
APPROVE	$PD < 0.10$	71.71%
REVIEW	$0.10 \leq PD < 0.18$	17.22%
REJECT	$PD \geq 0.18$	11.07%

The held-out population contains 30,873 APPROVE, 7,412 REVIEW and 4,767 REJECT decisions. Threshold logic, complete customer assignment and deterministic repeated prediction were validated.

7. Risk Segmentation

Risk band	Customers	Average PD	Observed default rate
Low Risk	18,987	0.0313	0.0289
Medium Risk	11,886	0.0711	0.0727
High Risk	7,412	0.1340	0.1218
Very High Risk	4,767	0.2441	0.2465

Observed default rates and average PD increase across the risk bands, providing monotonic risk ordering.

8. Expected Loss and Business Impact

Expected loss is calculated as $PD \times Exposure \times LGD$. The exposure variable is AMT_CREDIT and the analysis uses an LGD assumption of 0.45.

Portfolio measure	Final value
Customers	43,052
Total exposure	25,974,998,023.50
Average PD	0.0835
Total expected loss	905,828,407.72
Expected loss rate	3.4873%
High + Very High customer share	28.29%
High + Very High expected-loss share	55.96%
Very High expected-loss share	28.98%

The central business finding is that 28.29% of customers account for approximately 55.96% of expected loss, indicating substantial concentration of modeled portfolio risk.

9. Deployment

The final deployment package contains the saved preprocessor, LightGBM model, calibration model, feature metadata, decision configuration and model metadata. Reload tests produced identical model probabilities and calibrated PDs. The complete inference flow is: raw/engineered inputs → preprocessing → 226 model features → LightGBM raw PD → calibration → decision and risk bands.

10. Stress and Edge-Case Testing

Test	Status
Single customer	PASS
100-customer batch	PASS
Missing values	PASS
Extreme numerical values	PASS
Unseen categorical value	PASS
Deterministic repeated prediction	PASS
Invalid feature count protection	PASS

11. Governance and Audit

The final governance audit covers model performance, OOF/generalization, decision-engine configuration, business impact, deployment artifacts and stress testing. Governance completeness was recorded as 100% with an EXCELLENT grade.

Governance component	Status
Performance audit	Complete
OOF/generalization audit	Complete
Decision-engine audit	Complete
Business-impact audit	Complete

Governance component	Status
Deployment audit	Complete
Stress testing	Complete
Governance completeness	100%

12. Final Scorecard

Metric	Final result
Test ROC-AUC	0.7395
Test PR-AUC	0.2264
Brier Score	0.0691
OOF ROC-AUC	0.7340 ± 0.0018
Train ROC-AUC	0.8571
Train-test gap	0.1176
Customers evaluated	43,052
Expected loss	905,828,407.72
Expected loss rate	3.4873%
High/Very High customer share	28.29%
High/Very High expected-loss share	55.96%
Governance completeness	100%

13. Limitations

- The train-test ROC-AUC gap of 0.1176 indicates remaining overfitting.
- Test ROC-AUC of 0.7395 leaves room for stronger discrimination through model comparison and tuning.
- The decision thresholds are validated business rules; future work can optimize them against explicit financial costs.
- SHAP-based local and global explanations are identified as future work.
- Live production monitoring for drift, calibration degradation and performance is identified as future work.

14. Future Improvements

- Regularization and stronger validation to reduce the generalization gap.
- Alternative model and hyperparameter comparison.
- Additional domain-specific feature engineering.
- Cost-sensitive threshold optimization.
- SHAP-based explainability.
- Production drift, calibration and performance monitoring.

15. CV-Ready Project Bullets

- Built an end-to-end credit risk decision engine using LightGBM and probability calibration, achieving 0.7395 ROC-AUC and 0.2264 PR-AUC on 43K held-out customers.
- Engineered 106 raw/derived inputs into a 226-feature modeling pipeline and validated stability using 5-fold OOF evaluation (0.7340 ± 0.0018 ROC-AUC).

- Converted calibrated probability of default into APPROVE/REVIEW/REJECT decisions and four risk bands, validating monotonic risk ordering and decision-level default rates.
- Quantified portfolio-level financial risk using $PD \times Exposure \times LGD$, showing that High/Very High Risk customers represented ~28% of customers but ~56% of expected portfolio loss.
- Packaged preprocessing, model, calibration and decision artifacts and validated reproducible deployment through single-customer, batch, missing-value, unseen-category and edge-case stress tests.

16. Interview Explanation

I built a credit risk decision engine rather than stopping at a classification model. I used LightGBM to estimate probability of default, calibrated the probabilities, and converted them into APPROVE, REVIEW and REJECT decisions. I validated the model using a held-out test set and 5-fold OOF evaluation. I also connected predictions to expected financial loss using $PD \times Exposure \times LGD$, then packaged and stress-tested the complete deployment pipeline.

17. Completion Status

Component	Status
Data preparation	COMPLETE
Feature engineering	COMPLETE
Model training	COMPLETE
Cross-validation	COMPLETE
Test evaluation	COMPLETE
Probability calibration	COMPLETE
Decision engine	COMPLETE
Risk segmentation	COMPLETE
Expected-loss analysis	COMPLETE
Artifact packaging	COMPLETE
Deployment validation	COMPLETE
Stress testing	COMPLETE
Governance	COMPLETE
CV/Interview packaging	COMPLETE

Source note: Quantitative results are taken from the project's final scorecard, decision-engine summary, business-impact analysis, governance audit and completion package. No model results were changed for this report.